

Statistical Screening for Poisoned Machine-Learning Training Data

Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

github.com/poojakira/dataset-poisoning-detector - public, default branch main.

199

Tests passed

90.88%

Statement coverage

0.55

Cross-class F1 @ 5% poison

0.595

Cross-class F1 @ 10% poison

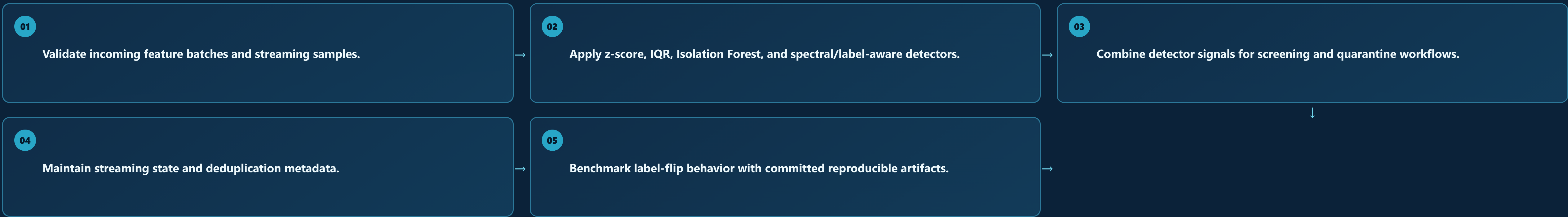
Contribution

An ingestion-boundary screening pipeline combining statistical outlier methods with a label-aware cross-class centroid detector that improves the committed synthetic label-flip benchmark from the old top-1 spectral baseline 0.08 / 0.23 / 0.3675 F1 to 0.55 / 0.595 / 0.6975 F1 at 5% / 10% / 20% contamination.

Verified Evidence

- 199 tests passed, 0 skipped.
- 90.88% statement coverage; CI gate is 90%.
- Ruff check/format and Pyright pass on the final branch; GitHub currently reports 0 open Code Scanning, Dependabot, and secret-scanning alerts.
- Committed results/spectral_benchmark.json records cross-class centroid F1:
- 0.55 at 5% contamination
- Cross-class F1 @ 20% poison: 0.6975
- Old top-1 spectral F1: 0.08 / 0.23 / 0.3675
- Feature-space ensemble F1: 0.0841 / 0.1421 / 0.2331

Architecture / Threat Flow



Flow labels come directly from the repository research brief. They describe the tested control path, not a universal security guarantee.

Methodology and Control Path

Validate incoming feature batches and streaming samples.

Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.

Combine detector signals for screening and quarantine workflows.

Maintain streaming state and deduplication metadata.

Benchmark label-flip behavior with committed reproducible artifacts.

Limitations and Residual Risk

- The benchmark is synthetic and separable; it is not proof of production detection rate.
- Clean-label, subtle backdoor, and very-low-rate poisoning remain difficult.
- Throughput is hardware/environment dependent; no fixed product SLA is claimed.
- Stateful streaming behavior requires deliberate partitioning/external state for horizontal scaling.

Reproducibility

```
git clone https://github.com/poojakira/dataset-poisoning-detector.git
cd dataset-poisoning-detector
git checkout 258a464abcc66bed2c9ccb538fa2f593140e750
python -m pip install -e ".[dev,realtime,kafka]"
pytest tests/ -q --cov=poison_detector --cov-report=term
python benchmark/cifar10_label_flip_benchmark.py
```

Expected current evidence: 199 passed, 90.88% coverage.